

CNN Motivation: Translation Equivalence

□ Translational equivalence

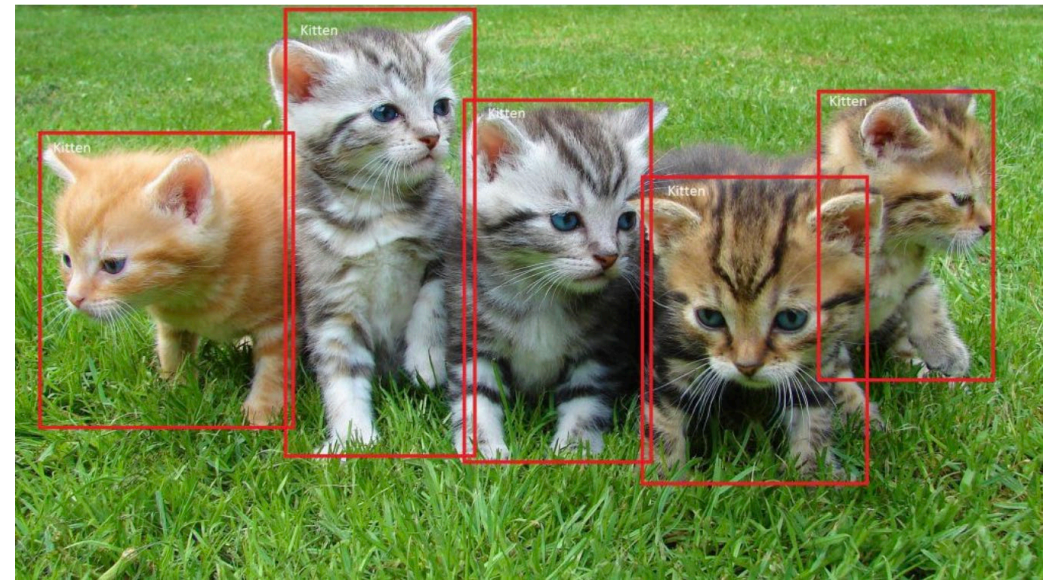
□ Intuitions: Same input at different location gives same output

□ Math details: $f(g(x)) = g(f(x))$

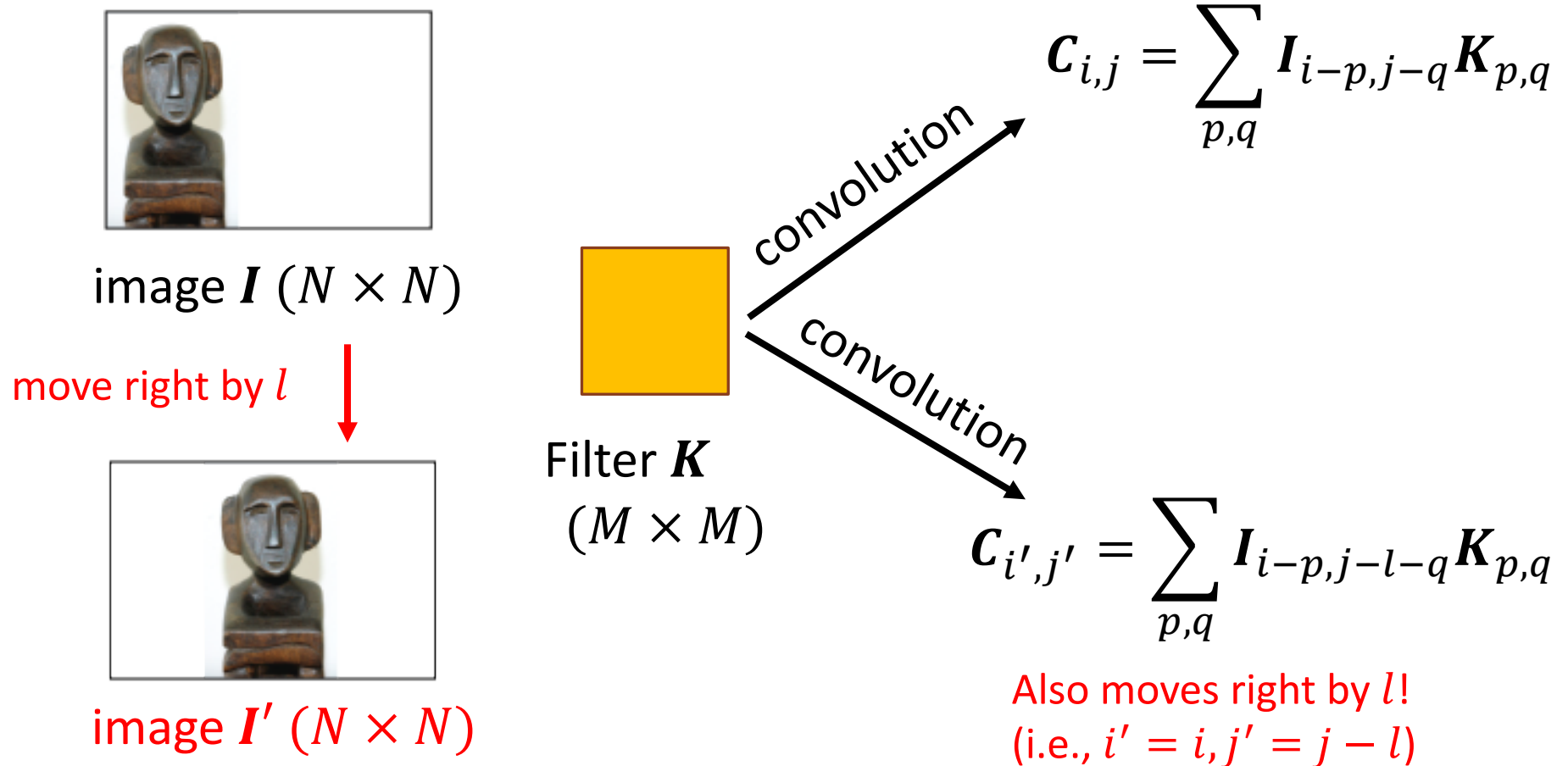
□ Examples

□ #1: a cat at the upper right corner and at the lower left corner of an image, will produce the same outputs

□ #2: “University of Illinois” at the start of the sentence and at the end of the sentence produce the same outputs



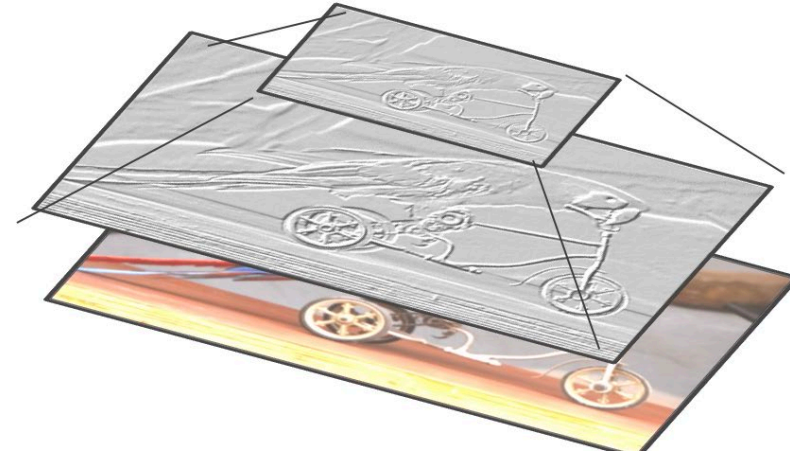
CNNs: Translation Equivalence



CNN: Pooling Layer

□ Pooling (Subsampling)

- Pool hidden units in the same neighborhood
 - Introduces invariance to local translations
 - Reduces the number of hidden units in hidden layer



Pooling
↑
Convolution
↑
Image

1	3	2	4
3	9	2	4
1	3	3	2
4	2	2	5

(a) Feature map

9	4
4	5

(b) Max pooling

4	3
2.5	3

(c) Mean pooling

CNN: Convolutional Layer

- ❑ Multiple kernels (organized in a tensor)
 - ❑ Applied separately to the input volume
 - ❑ Full in depth: producing a 2D feature map
 - ❑ Multiple feature maps are organized in a 3D tensor
- ❑ Followed by
 - ❑ nonlinear activation
 - ❑ pooling

